

Computer Networks

Lesson Objectives

By the end of this lesson, students should be able to:

- Explain what a computer network is and why it's important.
 - Differentiate between LAN, WAN, and MAN.
 - Identify common network devices and topologies.
 - Understand IP addressing and DNS.
 - Perform basic network troubleshooting commands.
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1. What is a Computer Network?

A **computer network** is a group of interconnected computers that communicate and share resources (such as files, printers, or internet access).

Examples:

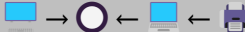
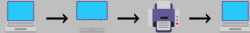
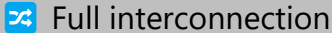
- Home Wi-Fi network.
 - Office or school computer lab network.
 - The Internet (global network of networks).
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2. Types of Networks

Type	Full Meaning	Coverage Area	Example
LAN	Local Area Network	Small area like a building or school	Office or lab network
MAN	Metropolitan Area Network	City or large campus	University network
WAN	Wide Area Network	Large geographical area	The Internet

3. Network Topologies

Network topology describes how devices are physically or logically connected.

Topology	Description	Diagram
Bus	All devices connected to a single central cable	
Star	All devices connected to a central hub/switch	
Ring	Devices connected in a circular path	
Mesh	Each device connected to many others	

Common Devices:

- **Switch** – Connects multiple computers in a LAN.
- **Router** – Connects multiple networks and directs data traffic.
- **Hub** – Basic connection device (less efficient than switch).

💡 4. IP Addressing & DNS

- **IP Address:** A unique number identifying each device on a network (e.g. 192.168.1.5).
- **DNS (Domain Name System):** Translates domain names (like google.com) into IP addresses.

⚙️ 5. Basic Troubleshooting Commands

Command	Description	Example
ping	Tests connection to another device	ping google.com
tracert	Shows path data takes to a destination	tracert google.com (Windows)
ipconfig	Displays local IP info	ipconfig

🧩 Activity

Task: Draw a simple home or office network diagram showing:

- Your **router**, **devices** (laptops, phones, printers), and **internet connection**.
- Label the **connections** (wired or wireless).

You can use tools like:

- **draw.io**, **Lucidchart**, or **pen and paper**.
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Summary

- Networks connect devices to share data and resources.
 - LAN, MAN, and WAN differ in **size** and **coverage**.
 - IP addresses and DNS ensure devices communicate correctly.
 - Basic tools like **ping** and **traceroute** help solve connection issues.
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Git Task

Create a new folder in your GitHub repository named **“networking”**.
Inside it:

1. Upload your network diagram (.png, .jpg, or .pdf).
2. Add a short description of your network setup in a .txt file.

Commit and push your changes to GitHub.

